

SAM 3 and Esri's Mask R-CNN on May 2026 Nearmap imagery

Two deep learning models, Meta's SAM 3 and Esri's Building Footprint Extraction – USA (a Mask R-CNN), outline buildings in 10 cm aerial imagery of Lewisville. This final report compares them across the whole city, in screenshots, on an Old Town sample and in a separate held-out test area, with each model's settings frozen by a fixed rule. Every score measures agreement with the city's building outlines collected in 2015 or earlier, not accuracy: outlines for an accuracy check were prepared but not reviewed.

- ⚠ Agreement, not accuracy
- ✓ All three city-wide runs complete
- ✓ Settings frozen, held-out test area scored once
- 🕒 Reviewed accuracy not done

Imagery: Nearmap, May 2026 · 29 September 2026 · Nearmap capture
2026-05, 10.16 cm · stored in EPSG:2276 (US survey feet), scored in
EPSG:26914

What the results show

0.95 · 0.91 · 0.87

Held-out test area, frozen settings

F1 on buildings of 20 m² and up in the held-out test area, scored once against its 2015 outlines: the city's Esri run 0.95, the local Esri run 0.91, SAM 3 0.87.

0.83 vs 0.74

Esri leads city-wide

City-wide F1 on buildings of 20 m² and up: the city's Esri run 0.83, SAM 3 0.74. Recall is close (0.85 against 0.82); SAM 3 draws more outlines the 2015 layer lacks.

91%

The two models mostly agree

91% of Esri's outlines have a matching SAM 3 outline. Matched to the 2015 outlines, SAM 3's fit slightly more tightly (median IoU 0.80 against 0.77).

0.83 vs 0.77

Esri's post-processing helps

The same Esri model run locally without its post-processing scores 0.77 city-wide on buildings of 20 m² and up. It finds more small structures but draws many more extra outlines.

4,923

Likely built since 2015

SAM 3 outlines with no matching 2015 outline that Esri's run also draws. Two independent models agreeing where the old layer has nothing points to new buildings, such as whole new subdivisions.

7,599

Small structures are missed

2015 outlines under 20 m² that neither the city's Esri run nor SAM 3 found, of 7,867. SAM 3 matched 267, the local Esri run 352 and the city's run 2.

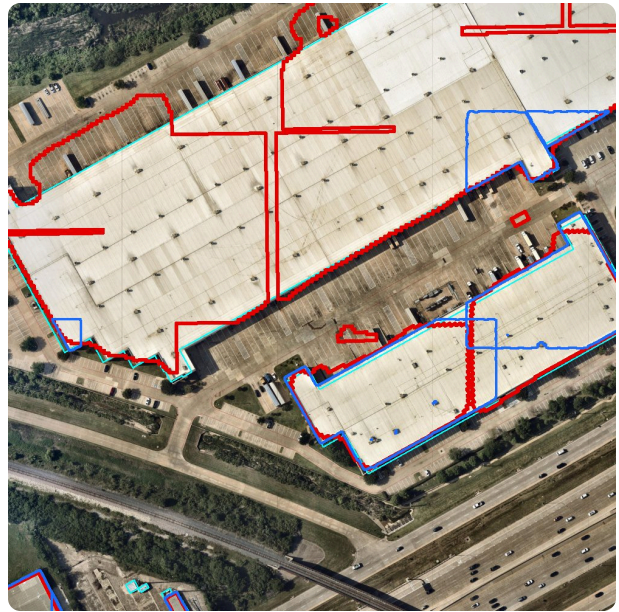
Screenshots: where the models agree and differ

Eight areas: two chosen earlier to show typical streets, and six picked by rule from the city-wide results, each the 300 m square with the most of one kind of difference.

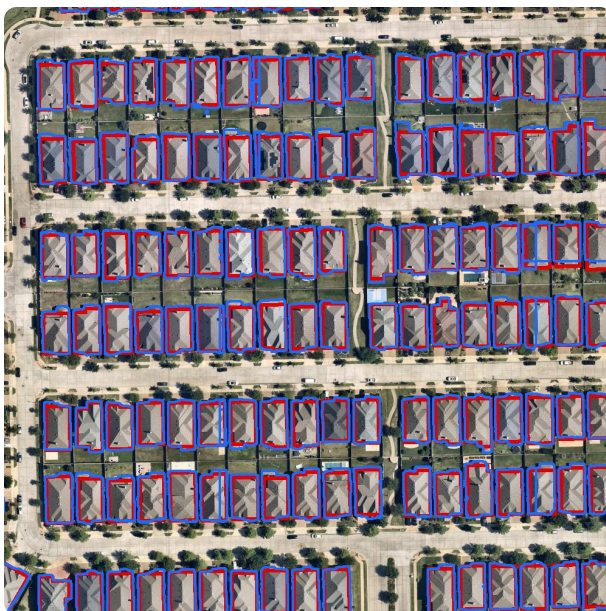
Esri, city run SAM 3: "building", 1536 px tiles, ≥ 0.7 City outlines, 2015 or earlier



Typical subdivision, 400 m square (33.0555° N, 96.9029° W). Both models outline almost every house, red and blue nearly on top of each other. A few blue lines cross roofs where SAM 3 drew overlapping outlines.



Distribution centre, 400 m square (33.0558° N, 96.9157° W). Esri outlines the largest roof but cuts it into blocks at its chip edges; SAM 3 finds only a corner of it and splits the smaller building where its tiles meet.



Built since 2015, 300 m square (33.0255° N, 96.8972° W). No 2015 outlines here, yet both models outline every house. This square has the most outlines that SAM 3 and Esri share with no 2015 counterpart (125).



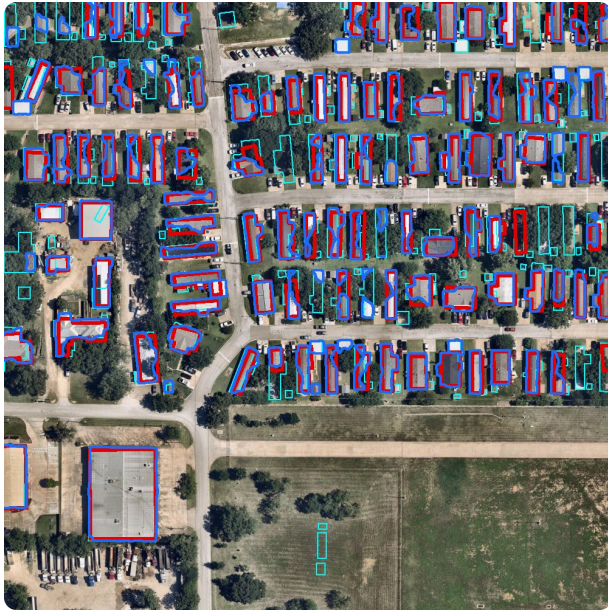
SAM 3's extra outlines, 300 m square (32.9923° N, 97.0167° W). A manufactured-home community beside a warehouse. This square has the most SAM 3 outlines that neither the 2015 layer nor Esri has (57), mostly small outbuildings and parts of homes. Many small cyan outlines are found by neither.



Where Esri finds more, 300 m square (33.0107° N, 96.9842° W). An apartment complex. Esri outlines the long carport and garage rows, drawn in red only, that SAM 3 mostly misses. This square has the most 2015 outlines found only by Esri (30).



Where SAM 3 finds more, 300 m square (33.0764° N, 97.0341° W). Older streets with back-yard garages and sheds. SAM 3 outlines more of the small outbuildings; this square has the most 2015 outlines found only by SAM 3 (24). Many small cyan outlines are still missed by both.



Sheds and small structures, 300 m square (33.0324° N, 96.9869° W). The most 2015 outlines under 20 m² missed by both models (61). The cyan outlines in the open field at the bottom mark buildings that have since gone.



Large buildings, 300 m square (33.0680° N, 97.0150° W). Industrial and storage buildings. Both models find the long roofs, but SAM 3 splits them with straight cuts where its tiles meet, and Esri's edges step at its chip boundaries.

Zoom in on a screenshot to see the full 1000 px image. The areas were picked by rule from the city-wide results and avoid the two review areas, so the reviews stay independent of the models. Imagery: Nearmap, May 2026.

City-wide: SAM 3 and Esri against the 2015 outlines

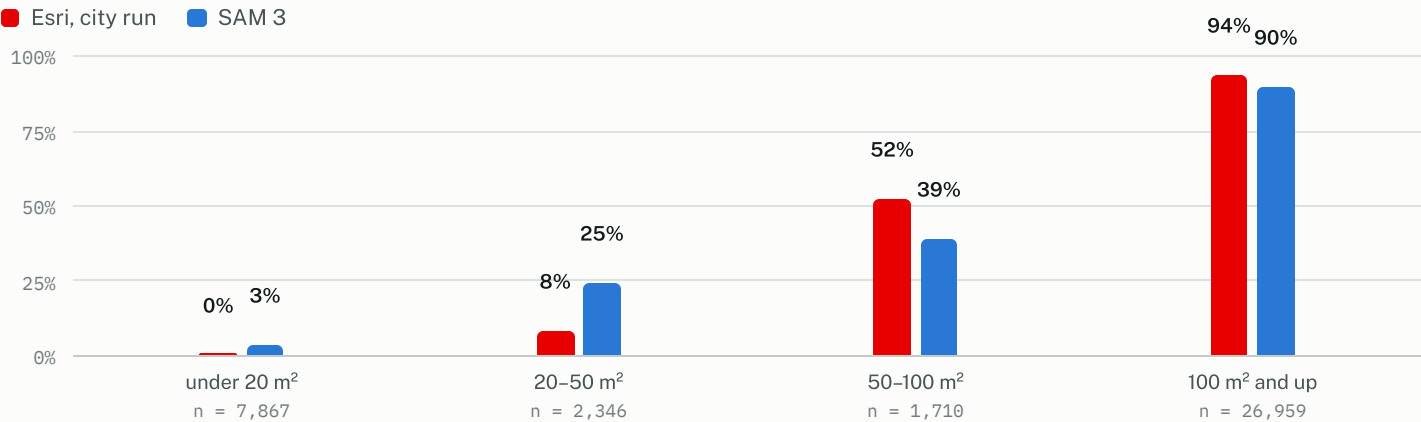
SAM 3 ran on all 6,538 tiles of 1536 px covering the city, with the settings the tuning rule later froze: the prompt “building” and a confidence threshold of 0.7. The same Esri model also ran locally over the whole city, in 150 chunks. All three went through the same cleanup and are scored against the city's 38,882 older outlines inside the city limits.

Scores inside the city limits

Method	Outlines	Matched (TP)	Extra (FP)	Missed (FN)	Precision	Recall	F1	F1, 20 m² and up	Median IoU
Esri, city run	32,608	26,502	6,106	12,380	0.81	0.68	0.74	0.83	0.77
SAM 3	37,906	25,758	12,148	13,124	0.68	0.66	0.67	0.74	0.80
Esri, local raw run	38,860	26,536	12,324	12,346	0.68	0.68	0.68	0.77	0.80

Extra outlines are not all errors: many are buildings built since 2015. On buildings of 20 m² and up, the tuning objective, the city's Esri run leads SAM 3 0.83 to 0.74, mostly on precision (0.82 against 0.68). Esri's local raw run is the same model without Esri's post-processing, with a confidence threshold of 0.9: F1 0.68, and 0.77 on buildings of 20 m² and up. It finds more small structures than either (352 under 20 m²) but draws many more extra outlines, so Esri's own post-processing in the city's run is worth keeping.

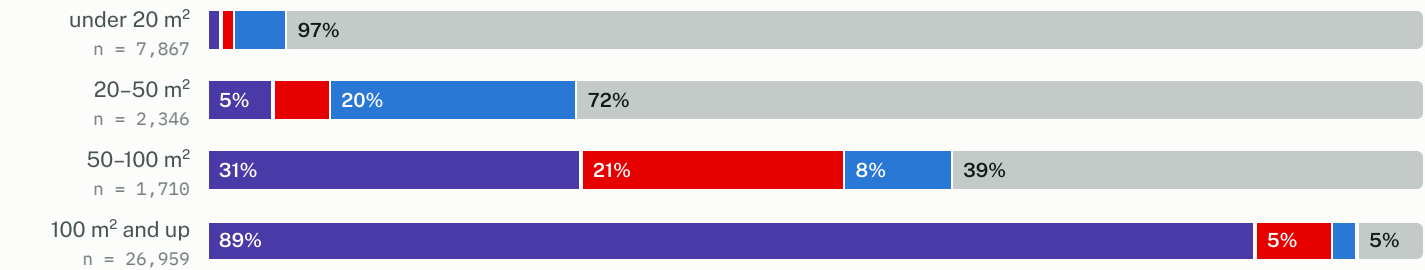
Share of 2015 outlines each model found, by building area



Building area	2015 outlines	Found by Esri, city run	Found by SAM 3
under 20 m²	7,867	2 (0.0%)	267 (3.4%)
20-50 m²	2,346	194 (8.3%)	577 (24.6%)
50-100 m²	1,710	896 (52.4%)	669 (39.1%)
100 m² and up	26,959	25,410 (94.3%)	24,245 (89.9%)

What happened to each 2015 outline

Found by both Only Esri Only SAM 3 Missed by both



Of the 38,882 2015 outlines, both models found 24,585, Esri alone 1,917, SAM 3 alone 1,173 and neither 11,207. 7,599 of those missed by both are under 20 m²; some others have been demolished since 2015. Adding the local Esri run, 10,463 outlines are missed by all three, 7,359 of them under 20 m².

Building area	Found by both	Only Esri	Only SAM 3	Missed by both
under 20 m²	1	1	266	7,599
20-50 m²	107	87	470	1,682
50-100 m²	530	366	139	675
100 m² and up	23,947	1,463	298	1,251

SAM 3 against Esri, without the 2015 outlines

29,567

91%

78%

0.81

4,923

Matched pairs

Esri outlines SAM 3 also drew

SAM 3 outlines Esri also drew

Median IoU between them

Likely built since 2015

Matched to each other at IoU ≥ 0.5, the same rule as for the 2015 outlines. Of SAM 3's 12,148 outlines with no 2015 match, 4,923 match an Esri outline; those are counted here as likely built since 2015.

Old Town sample: the untuned baseline

A 700 m square around Old Town, with 319 of the 2015 outlines. Each method ran at its default settings, with the same cleanup and scoring for all.

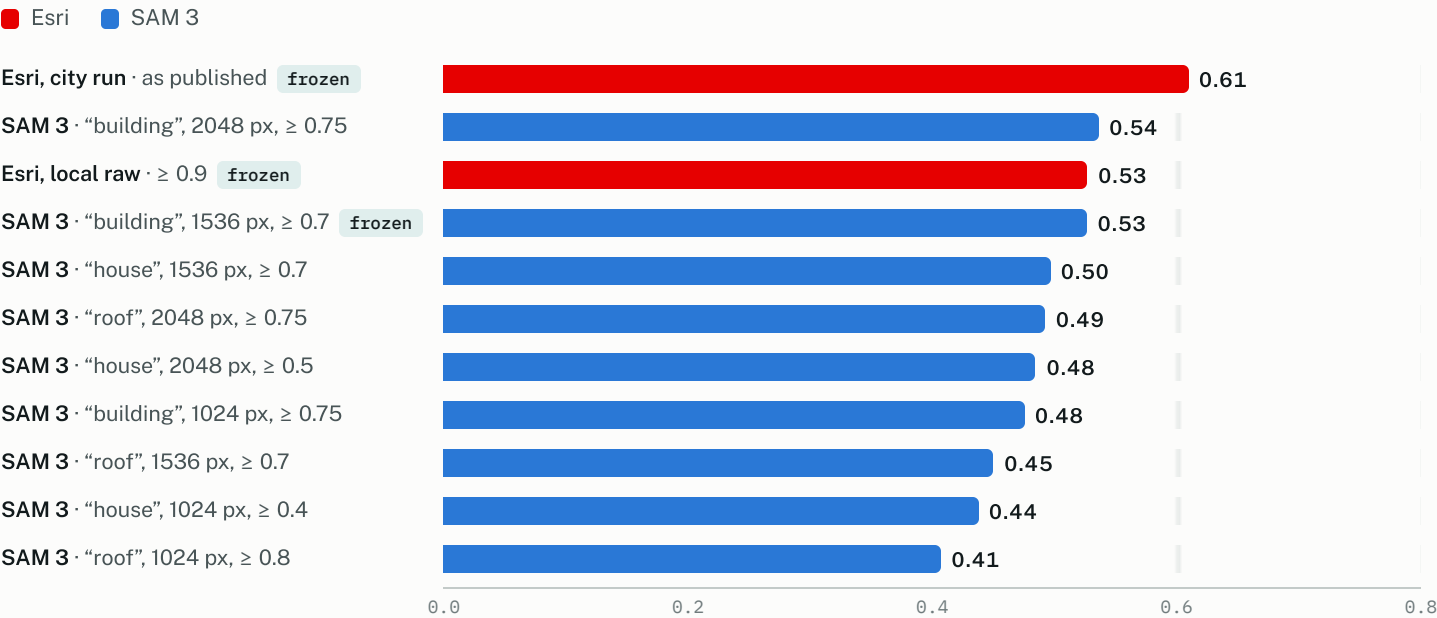
Method	Matched (TP)	Extra (FP)	Missed (FN)	Precision	Recall	F1	Median IoU
SAM 3	142	287	177	0.33	0.45	0.38	0.74
Esri, city run	135	63	184	0.68	0.42	0.52	0.75
Esri, local raw	158	381	161	0.29	0.50	0.37	0.74
Method	Recall ≥ 50 m²		Precision ≥ 50 m²		Recall < 50 m²		
SAM 3	0.63		0.39		0.10		
Esri, city run	0.64		0.69		0.01		
Esri, local raw	0.71		0.36		0.10		

Per building, out of 319 outlines: 109 found by all three methods, 38 by two, 32 by only one, and 140 by none. 96 of those 140 are under 50 m².

Tuning in Old Town

The tuning objective, fixed in advance, is F1 at IoU ≥ 0.5 on buildings of 20 m² and up in the Old Town square. With no reviewed outlines available, the final tuning used the 2015 outlines (Amendment 1 to the protocol), and the confidence threshold ranges were extended because the best values sat at the top of the original ranges. 176 variants in all; regularization never changed a score, so the charts show it off.

Best F1 for each setting



Method	Setting	Precision	Recall	F1	Median IoU	Recall under 20 m ²
Esri city run	as published	0.69	0.55	0.61	0.75	0.00
SAM 3	building, 2048 px, ≥ 0.75	0.60	0.48	0.54	0.76	0.00
Esri local	≥ 0.9	0.56	0.50	0.53	0.76	0.03
SAM 3	building, 1536 px, ≥ 0.7	0.53	0.52	0.53	0.75	0.03
SAM 3	house, 1536 px, ≥ 0.7	0.60	0.42	0.50	0.75	0.03
SAM 3	roof, 2048 px, ≥ 0.75	0.54	0.46	0.49	0.75	0.01
SAM 3	house, 2048 px, ≥ 0.5	0.49	0.48	0.48	0.74	0.04
SAM 3	building, 1024 px, ≥ 0.75	0.48	0.47	0.48	0.74	0.01
SAM 3	roof, 1536 px, ≥ 0.7	0.41	0.50	0.45	0.75	0.03
SAM 3	house, 1024 px, ≥ 0.4	0.38	0.51	0.44	0.73	0.14
SAM 3	roof, 1024 px, ≥ 0.8	0.38	0.43	0.41	0.74	0.04

SAM 3 F1 by prompt, tile size and confidence threshold

Prompt	Tile size	≥ 0.3	≥ 0.4	≥ 0.5	≥ 0.6	≥ 0.7	≥ 0.75	≥ 0.8	≥ 0.85	≥ 0.9
“building”	1024 px	0.38	0.41	0.43	0.44	0.46	0.48	0.47	0.39	0.19
“building”	1536 px	0.43	0.46	0.46	0.49	0.53	0.52	0.51	0.47	0.28
“building”	2048 px	0.43	0.46	0.47	0.51	0.53	0.54	0.50	0.44	0.26
“house”	1024 px	0.43	0.44	0.43	0.43	0.42	0.42	0.41	0.36	0.17
“house”	1536 px	0.44	0.47	0.46	0.47	0.50	0.47	0.45	0.38	0.18
“house”	2048 px	0.44	0.47	0.48	0.48	0.45	0.44	0.41	0.36	0.17
“roof”	1024 px	0.29	0.31	0.34	0.35	0.36	0.37	0.41	0.40	0.29
“roof”	1536 px	0.34	0.37	0.39	0.41	0.45	0.45	0.45	0.42	0.33
“roof”	2048 px	0.36	0.40	0.42	0.44	0.47	0.49	0.46	0.39	0.26

lower F1  higher F1

Bigger tiles and higher confidence thresholds help up to about 0.75; above that, recall falls faster than precision rises. SAM 3's best, 2048 px at 0.75 (F1 0.54), is within 0.01 of 1536 px at 0.7 (0.53), which the rule prefers as closer to the defaults. The local Esri run's confidence threshold sweep: ≥ 0.5: F1 0.43 · ≥ 0.6: F1 0.46 · ≥ 0.7: F1 0.49 · ≥ 0.8: F1 0.51 · ≥ 0.9: F1 0.53 · ≥ 0.95: F1 0.48.

Frozen settings and the held-out test area

Each model's settings were frozen by the rule in the protocol, and they match what the city-wide runs used. The held-out test area, a 700 m square chosen by a seeded random rule, was then scored once against its 2015 outlines, from the city-wide outputs.

Settings frozen by the rule

Method	Frozen settings	Old Town F1	Why
Esri, city run	as published, regularization off	0.61	only cleanup varies; regularization changed nothing
SAM 3	“building”, 1536 px tiles, confidence threshold 0.7, regularization off	0.53	within 0.01 of the best (2048 px, 0.75: 0.54) and closer to the defaults
Esri, local raw run	confidence threshold 0.9, regularization off	0.53	highest F1; 0.95 scores lower

Held-out test area, scored once: buildings of 20 m² and up

Method	Matched (TP)	Extra (FP)	Missed (FN)	Precision	Recall	F1	Median IoU	F1, all sizes
Esri, city run	445	16	26	0.97	0.94	0.95	0.76	0.83
SAM 3	430	92	41	0.82	0.91	0.87	0.75	0.76
Esri, local raw run	422	30	49	0.93	0.90	0.91	0.79	0.80

Agreement is far higher than in Old Town, where much has been built or rebuilt since 2015, but the order is the same. SAM 3 finds nearly as many of the 2015 buildings (recall 0.91) but draws more outlines they lack (precision 0.82). Small structures are missed everywhere: at most 2 of the 136 outlines under 20 m².

Runtime on this laptop

An RTX 4050 laptop GPU with 6 GB. SAM 3 used up to 96% of the GPU's memory, close to the limit but stable through the city-wide run.

Run	Area or tiles	Time	City-wide
SAM 3, 1024 px tiles	90 tiles (Old Town)	105 s (0.82 s per tile)	about 3.3 h, estimated
SAM 3, 1536 px tiles	6,538 tiles (whole city)	2 h 41 min (median 1.08 s of model time per tile)	measured
Esri, local raw run	0.64 km² (Old Town)	4.9 min	estimate for the city: about 15 h
Esri, local raw run, whole city	150 chunks of about 1.1 km	8 h 46 min of model time	measured

How this was run

Imagery	Nearmap mosaic, May 2026, 10.16 cm, 21.1 gigapixels, clipped to the city limits. Its GeoTIFF declared a local CRS in metres; it is EPSG:2276 in US survey feet, corrected through a VRT and checked against the city limits.
Reference	The City of Lewisville's building outlines, collected in 2015 or earlier (38,924 outlines, public). Aligned with the imagery to within about 0.4 m.
SAM 3	Meta's SAM 3 with a text prompt, through SamGeo 1.4.2, at a pinned commit. Tuning used 1024–2048 px tiles; the city-wide run used 1536 px tiles overlapping by 192 px, the prompt “building” and a confidence threshold of 0.7.
Esri, city run	Building Footprint Extraction – USA (Mask R-CNN) through ExtractFeaturesUsingAIModels with default settings, including Esri’s post-processing. Published by the city.
Esri, local raw run	The same model through DetectObjectsUsingDeepLearning, without non-maximum suppression (NMS), keeping confidence scores. City-wide it runs in 150 overlapping chunks of about 1.1 km; each chunk keeps only the outlines centred in its own core.
Scoring	Identical for every method: EPSG:26914, 4 m ² minimum, duplicate suppression, one-to-one matching at IoU ≥ 0.5, and buildings crossing an area’s edge excluded.

What would come next

1. For accuracy: review the Old Town and test-area outlines against the 2026 imagery (the review packages are ready), then score the frozen settings against them. No tuning is needed again.
2. For small structures: fine-tune a model on labelled structures under 20 m²; none of these methods finds them.
3. For large buildings: merge outlines across tile seams, or use larger tiles in commercial and industrial areas.